# Structured Computer Architecture
# Chapter 2.11: Processors
# Third-Order Systems (3-OS)

An Order-Based Approach

John Glossner and Gheorghe M. Ștefan

Based on the textbook

September 14, 2025

# Outline

# Learning Objectives

By the end of this chapter, you will be able to:

- Understand third-order systems (3-OS) and the third feedback loop
- Analyze the structure of elementary processors
- Design RALU (Registers with ALU) components
- Understand CROM (Control ROM) operation
- Explain micro-programming concepts
- Recognize context-free language processing capabilities

# Evolution to 3-OS: The Third Loop

- **Previous Systems Review**:
  - **0-OS**: No loops, combinational circuits
  - **1-OS**: One loop, memory capability
  - **2-OS**: Two loops, autonomous behavior
- **3-OS Innovation**: Third feedback loop
  - Can be closed through combinational circuit
  - Can be closed through memory circuit
  - **Focus**: Closed through automaton over automaton
  - Enables processor functionality
- **Key Achievement**:
  - Complex computational behavior
  - Context-free language recognition
  - Elementary processor implementation

# Context-Free Languages and Processors

- **Language Hierarchy**:
  - **Type 3 (Regular)**: Recognized by finite automata
  - **Type 2 (Context-Free)**: Require more powerful systems
  - **Type 1 (Context-Sensitive)**: Even more complex
  - **Type 0 (Unrestricted)**: Turing machine equivalent
- **3-OS Capability**:
  - Recognizes context-free languages
  - Processes nested structures
  - Handles recursive patterns
  - Enables programming language parsing
- **Practical Applications**:
  - Compiler design
  - Expression evaluation
  - Nested procedure calls
  - Balanced parentheses checking

# Beyond Finite Automata Limitations

- **Finite Automaton Limitation**:
  - Cannot recognize $\{1^n0^n | n > 0\}$
  - Fixed number of states independent of $n$
  - No counting capability
- **Counter-Expanded Automaton (CEA)**:
  - Finite automaton + reversible counter
  - Counter provides zero flag to automaton
  - Creates 3-OS system
  - Enables context-free language recognition
- **Enhanced Capabilities**:
  - Counting and comparison operations
  - Stack-like behavior simulation
  - Nested structure processing
  - Context-free grammar parsing